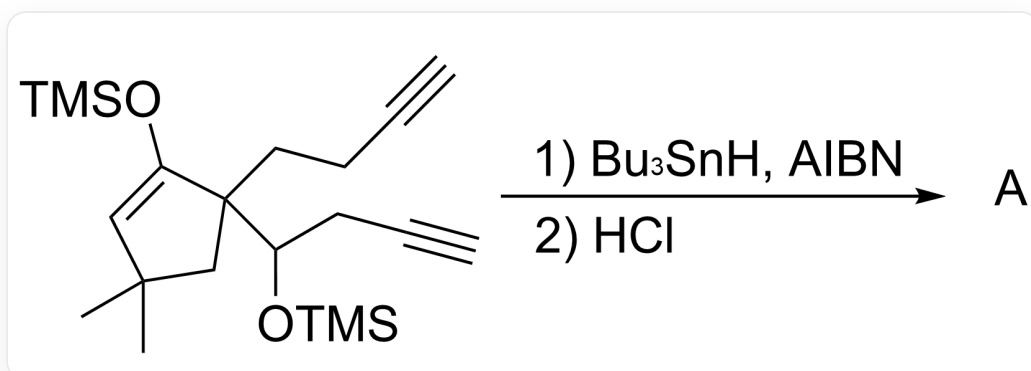
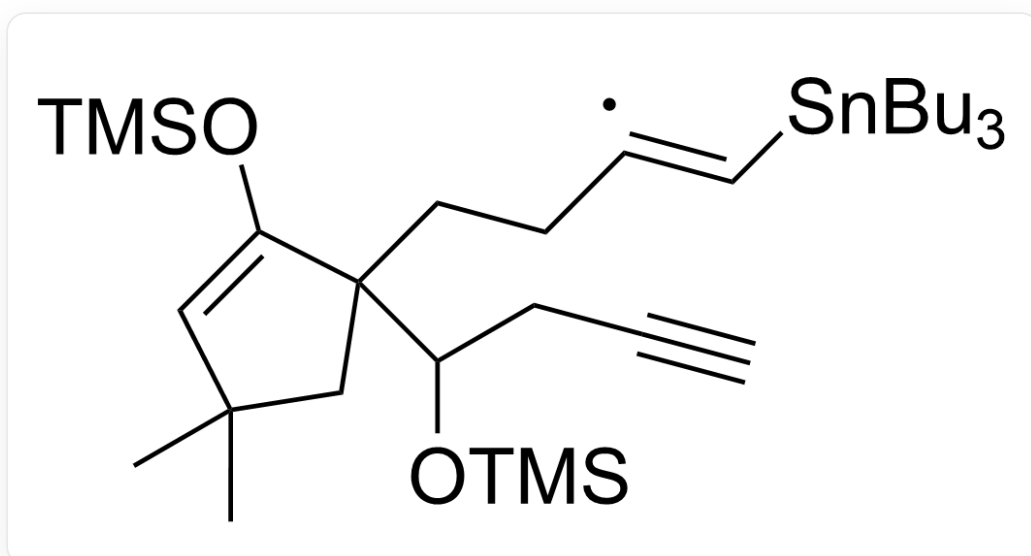


题目

自由基重排反应在有机合成中有着广泛的应用

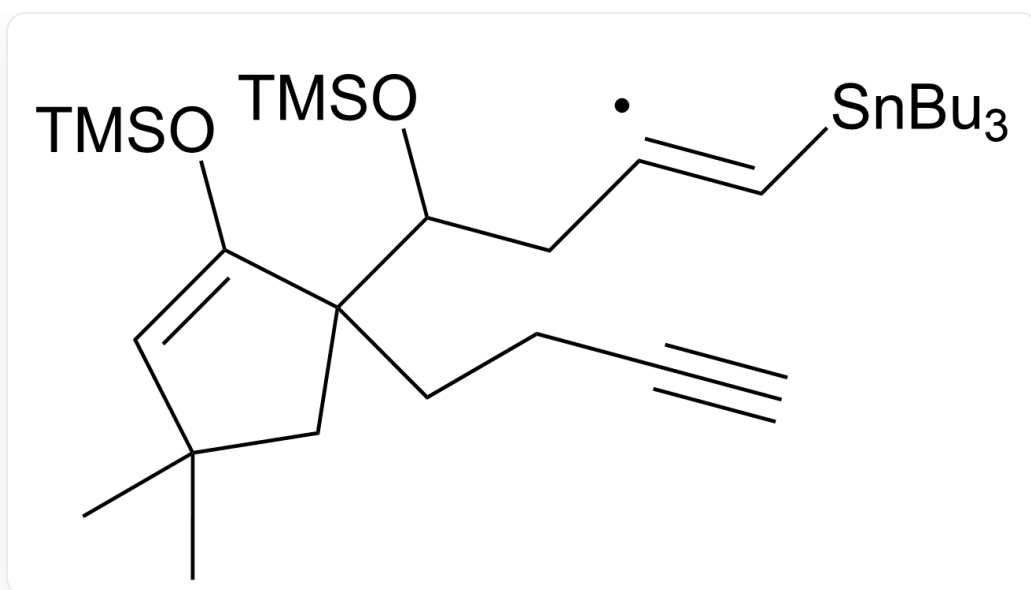


C#CCCC1(C(O[Si](C)(C)C)CC#C)CC(C)(C)C=C1O[Si](C)(C)C>[Bu_3SnH].[AIBN]>[A], A是反应产物，反应完成后加入 HCl



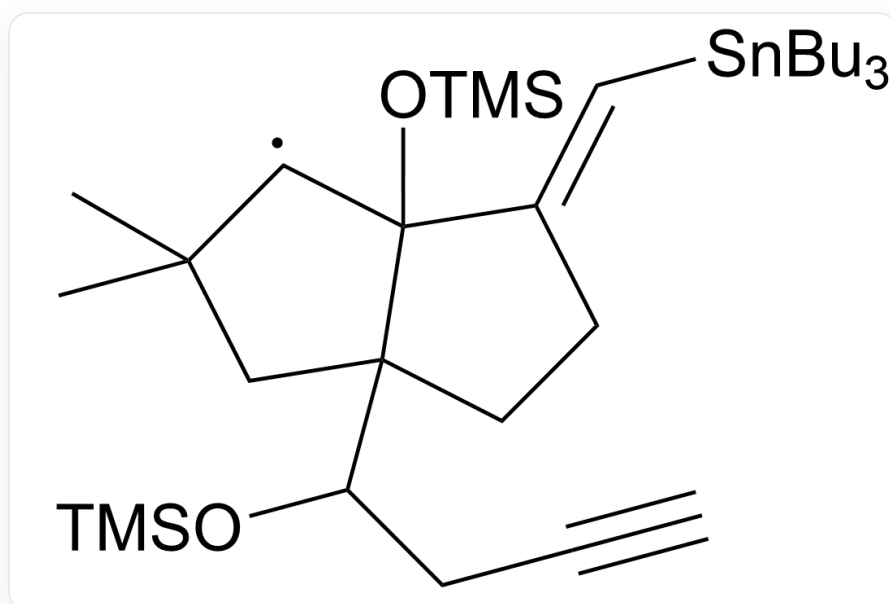
C#CCCC1(CC(C)(C=C1O[Si](C)(C)C)C(C/C([X])=C/[Sn](CCCC)(CCCC)CCCC)O[Si](C)(C)C,X为单电子,
选项1

选项1



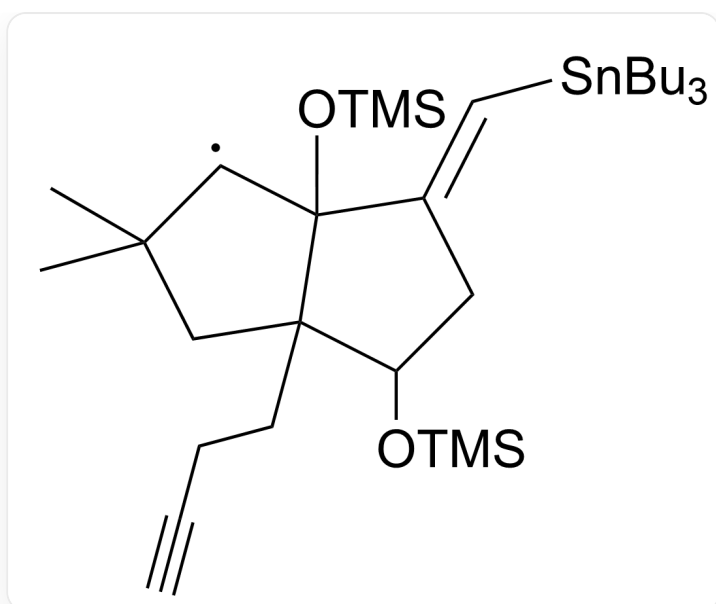
C#CCCC1(CC(C)(C=C1O[Si](C)(C)C)C)C(C/[X])=C/[Sn](CCCC)(CCCC)CCCC)O[Si](C)(C)C,X为单电子,
选项2

选项2



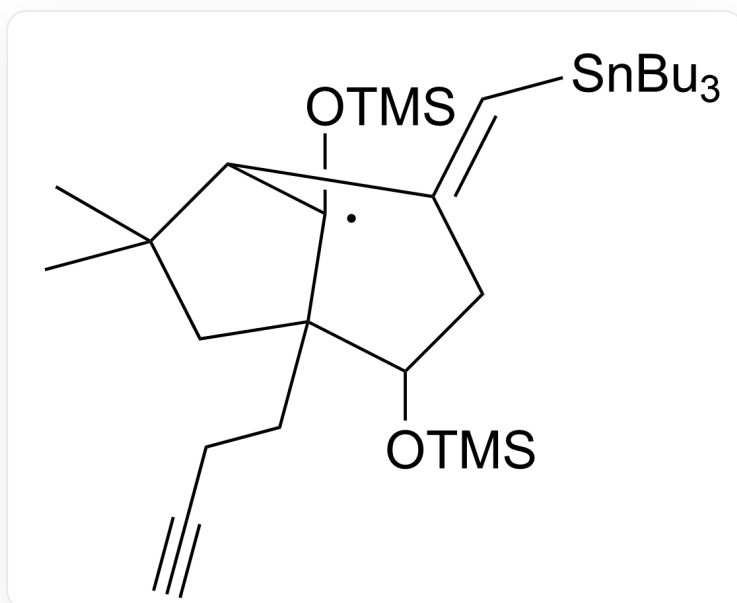
C#CCC(C1(CC(C)(C(C1/2O[Si](C)(C)C)[X])C)CCC2=C[Sn](CCCC)(CCCC)CCCC)O[Si](C)(C)C,X是单电
子,选项3

选项3



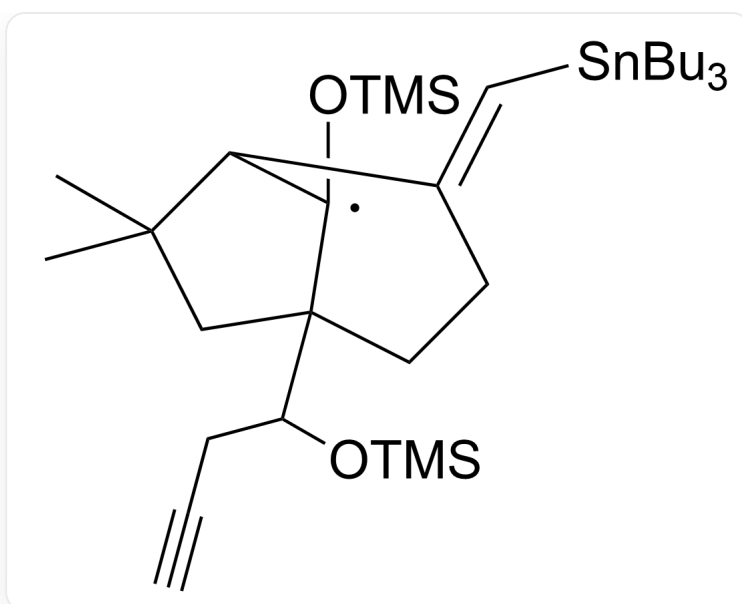
CC1(C([X])C2(C(CCC#C)(C1)C(C/C2=C\[Sn](CCCC)(CCCC)CCCC)O[Si](C)(C)C)O[Si](C)(C)C)C,X是单电子, 选项4

选项4



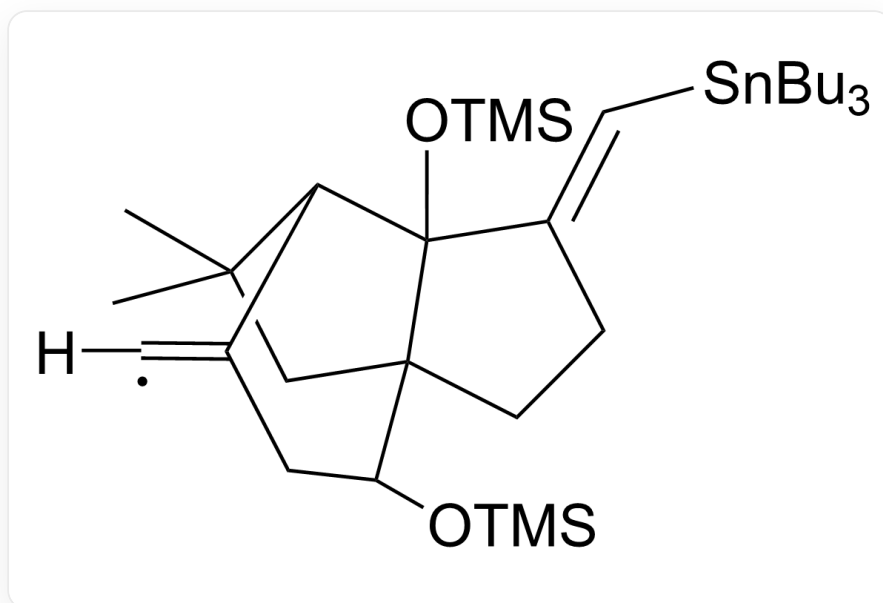
CC1(C2C(C(CCC#C)(C1)C(C/C2=C\[Sn](CCCC)(CCCC)CCCC)O[Si](C)(C)C)([X])O[Si](C)(C)C)C,X是单电子,选项5

选项5



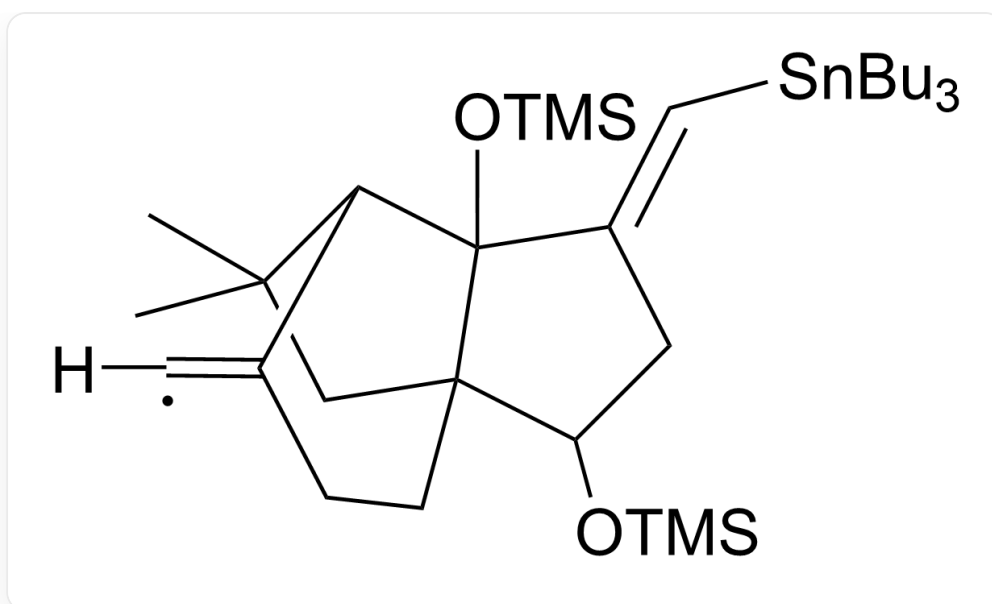
CC1(C2C(C(C(CC#C)O[Si](C)(C)C)(C1)CC/C2=C\[Sn](CCCC)(CCCC)CCCC)([X])O[Si](C)(C)C)C,X是单电子,选项6

选项6



CC1(C2C3(C(C(C/C2=C([X])\[H])O[Si](C)(C)C)(C1)CC/C3=C\[Sn](CCCC)(CCCC)CCCC)O[Si](C)(C)C)C,X是单电子,选项7

选项7



CC1(C2C3(C(CC/C2=C([X])\[H])(C1)C(C/C3=C\[Sn](CCCC)(CCCC)CCCC)O[Si](C)(C)C)O[Si](C)(C)C)C,X

是单电子,选项8

选项8

不考虑对映异构的情况下, 请选出选项中所有可能的中间体

A. 其他选项均不正确

B. 选项1、3、5

C. 选项2、4、5

D. 选项1、3、7

E. 选项2、4、8

F. 选项1、3、6

G. 选项2、5

H. 选项2、5、6

I. 选项1、4、6

答案

正确答案: **F**

详细解析

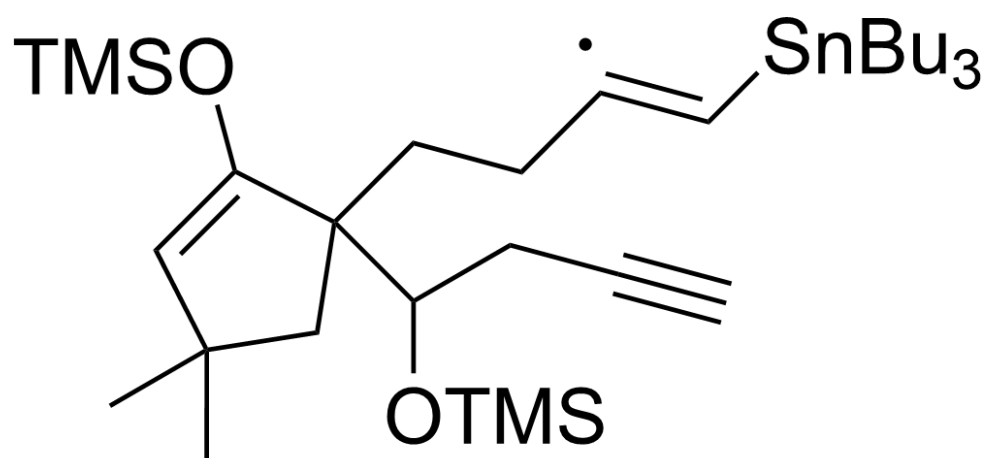
在 γ 位引入 $-OTMS$ 吸电子基团的碳自由基相对较为不稳定

CHECKPOINT

1 PTS

在 γ 位引入 $-OTMS$ 吸电子基团的碳自由基相对较为不稳定

首先形成中间体



C#CCCC1(CC(C)(C=C1O[Si](C)(C)C)C(C/C([X])=C/[Sn](CCCC)(CCCC)CCCC)O[Si](C)(C)C,X为单电子

CHECKPOINT

1 PTS

首先形成中间体：C#CCCC1(CC(C)(C=C1O[Si](C)(C)C)C(C/C([X])=C/[Sn](CCCC)(CCCC)CCCC)O[Si](C)(C)C, X为单电子

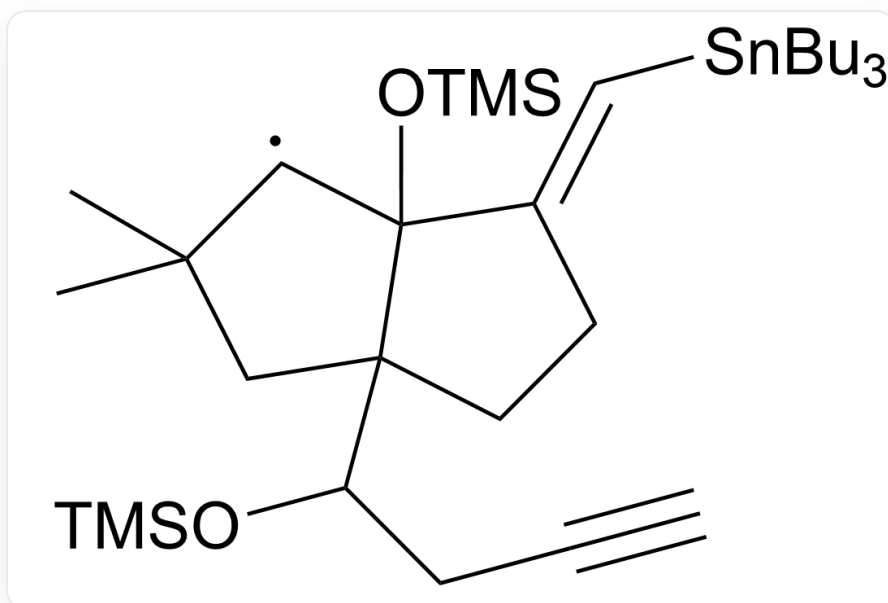
5-exo-trig的反应速率较快

CHECKPOINT

1 PTS

5-exo-trig的反应速率较快

因此优先生成中间体



CC1(C([X])C2(C(CCC#C)(C1)C(C/C2=C/[Sn](CCCC)(CCCC)CCCC)O[Si](C)(C)C)O[Si](C)(C)C, X是单电子

CHECKPOINT

1 PTS

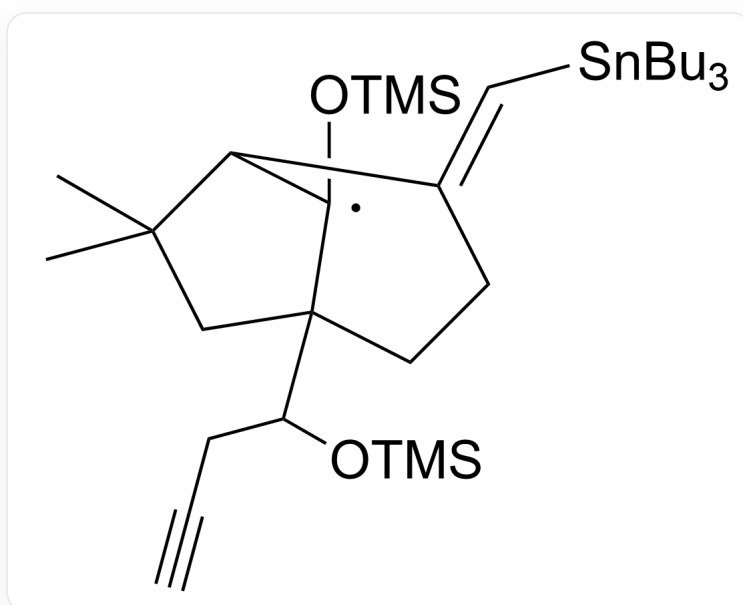
接下来优先形成中间体：CC1(C([X])C2(C(CCC#C)(C1)C(C/C2=C\[Sn](CCCC)(CCCC)CCCC)O[Si](C)(C)C)O[Si](C)(C)C)C, X是单电子

然后再经过一步快速的1,2重排反应得到被氧的n电子稳定的碳自由基

CHECKPOINT

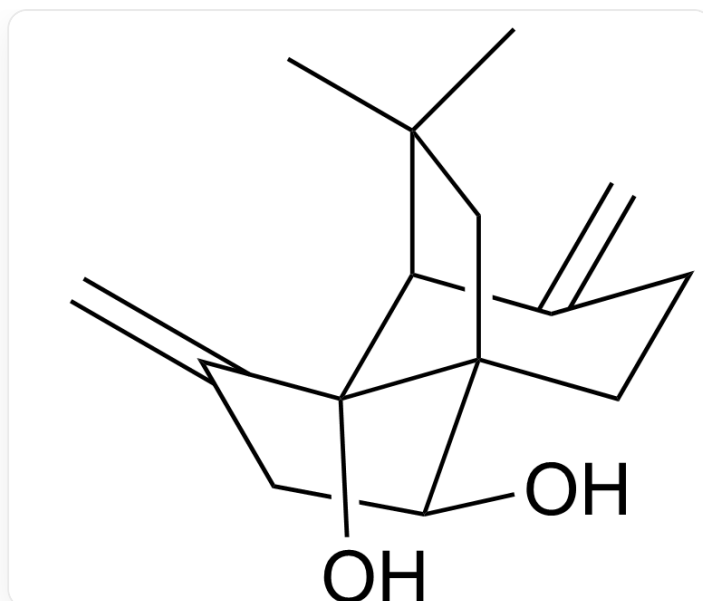
1 PTS

然后再经过一步快速的1,2重排反应得到被氧的n电子稳定的碳自由基



CC1(C2C(C(C(CC#C)O[Si](C)(C)C)(C1)CC/C2=C\[Sn](CCCC)(CCCC)CCCC)([X])O[Si](C)(C)C)C, X是单电子

最后经得到的产物A为



CC1(C)[C@@H]2[C@@](C(C[C@@H]3O)=C)(O)[C@]3(C1)CCC2=C